HARIPRASAD BATHINI SANKARAN

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DATA ENGINEER

Summary

Data Engineer with 5+ years building scalable batch and streaming pipelines that turn raw, multi-terabyte data into analytics-ready assets teams rely on daily. Track record of cutting processing latency, unifying ingestion across 50+ sources, and shipping reusable frameworks that shorten new-feed delivery from weeks to days. Deep Databricks and PySpark expertise across Azure and AWS, with hands-on domain experience in cybersecurity telemetry and healthcare (FHIR) data.

Skills

Data Engineering: Databricks, PySpark, Apache Spark, Spark SQL, Structured Streaming, Python, SQL, Delta Lake, ETL/ELT, Data Modeling (Dimensional / Star Schema), Medallion Architecture, Unity Catalog, Databricks Workflows, Autoloader, Data Quality & Governance
Cloud & DevOps: Azure (Data Factory, ADLS Gen2, SQL Server), AWS (Lambda, S3), Terraform, Git/GitHub, CI/CD
Streaming & Ingestion: Apache Kafka, Cribl, REST API Ingestion, Autoloader
Business Intelligence: Power BI, Tableau, Dashboards, Operational Analytics, Reporting
Programming: Python, C++, Java
Familiar with: Snowflake, Apache Airflow, dbt, MLflow
Tools: VS Code, Cursor, GitHub Copilot, Genie AI, Unix/Bash

Certification

- **Databricks Certified Data Engineer Professional (Link)** — *Dec 2025*
- **Azure Data Fundamentals (DP-900) (Link)** — *Mar 2022*
- **Azure Fundamentals (AZ-900) (Link)** — *Jul 2021*

Work Experience

Student Associate - Data Analyst | ***Bearcat Package Center*** | ***University of Cincinnati*** — **Jan 2026 – Present**

- Provided supervisior with one reliable source of truth by building and maintaining ETL pipelines that centralize package center data into a single reporting database, replacing scattered manual records.
- Delivered Power BI dashboards that turn raw package center metrics into at-a-glance operational views, speeding day-to-day decisions for the team.
- Reduced report generation time by optimizing SQL queries and data models, giving stakeholders faster access to the metrics they act on.
- Reduced manual data verification effort with automated validation scripts that catch integrity and consistency issues across sources before they reach reports.

Data Engineer | ***Comcast Corporation*** — **Aug 2023 – Aug 2025**

- Cut new data source onboarding effort 40% by building reusable pipeline templates the whole team adopted, delivering new analytics feeds in days instead of weeks.
- Reduced data processing latency 30%+ by orchestrating Databricks Workflows and Delta Live Tables with Terraform provisioned, auto-scaling compute landing in Unity Catalog managed Delta tables.
- Unified logs from 50+ security tools into a single analytics layer, automating ingestion with Cribl, Airflow, and AWS Lambda (JSON → S3) and ending fragmented, tool by tool data access for analysts.
- Gave threat-detection teams correlated, query ready data across global networks by building Databricks pipelines that process multi-terabyte telemetry from firewalls, endpoints, and detection systems.
- Enabled threat hunting teams to spot suspicious activity within minutes through realtime Structured Streaming and Kafka pipelines for cybersecurity logs.
- Sped threat investigations for downstream analysts by engineering a Bronze → Silver → Gold medallion architecture that delivers clean, structured, analytics-ready security data.
- Improved incident response accuracy by partnering with security analysts and data scientists to design log schemas, enrich threat intelligence feeds, and operationalize anomaly-detection models (Databricks MLflow).

Data Engineer | ***Infosys Ltd. (Client: Microsoft Corporation)*** — **Jun 2021 – Aug 2023**

- Reduced new data source integration time by 50% organization wide by building a reusable Python REST API ingestion notebook that other teams adopted for external data onboarding.
- Cut daily batch runtime by 30+ minutes and improved performance 15% by converting legacy SQL/T-SQL stored procedures into modular PySpark ETL frameworks.
- Improved data consistency and refresh accuracy 40%+ by engineering large-scale ETL in Azure Databricks and ADF that ingests and aggregates multi-source data into ADLS Gen2.
- Accelerated transformations across billions of records by tuning PySpark partitioning, caching, and shuffle strategies for high-performance distributed jobs.

- Increased pipeline scalability and maintainability with a metadata-driven ADF ingestion framework automating source-to-target mapping, schema validation, and dynamic orchestration.
- Enabled self-service analytics and executive dashboards by designing dimensional (fact/dimension) data models for reliable, decision-ready reporting.

Software Engineer - Data | ***Zoho Corporation Pvt. Ltd.*** **Jan 2020 – Dec 2020**

- Improved report accuracy and cut data-request turnaround by authoring and optimizing complex SQL for BI reports used by product and marketing teams.
- Reduced manual effort in recurring workflows with Python scripts for data cleaning, preprocessing, and file automation.
- Supported automation of key business metrics and dashboards by building and maintaining SQL-driven pipelines for internal reporting platforms.

Projects

Security Log Analytics and Threat Detection Platform (Azure) | *Personal Project* **Link**

- Built an end-to-end security-telemetry analytics platform on ADF and Databricks, processing firewall, endpoint, and network logs from ADLS Gen2 into SIEM-ready datasets via a Bronze → Silver → Gold medallion architecture.
- Automated infrastructure and scheduling with Terraform, Azure DevOps, and Apache Airflow for version-controlled, SLA-compliant, fully auditable production runs.
- Delivered Power BI and Synapse dashboards surfacing real-time threat anomalies, top attack sources, and detection metrics for cybersecurity teams.

Hybrid Healthcare Data Engineering Pipeline (FHIR → Medallion → RBAC) | *Personal Project* **Link**

- Engineered a Bronze → Silver → Gold pipeline ingesting synthetic FHIR R4 bundles with PySpark, using dynamic resource fan-out to auto-discover and build dedicated Silver tables for Claims, Patients, Encounters, and Conditions.
- Combined batch (PySpark) and real-time streaming (Kafka on Docker → Structured Streaming) to simulate concurrent hospital feeds across 50 partitioned data folders.
- Modeled a Kimball star schema (2 fact, 9 dimension tables) and a PostgreSQL serving layer with RBAC, column-level security, and dynamic masking enforcing field-level PHI restrictions across Admin and Clinical Analyst roles.

Enterprise RAG Agent: Agentic Document Q&A with LangChain and Databricks

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline with LangChain and Azure OpenAI for grounded, context-aware Q&A over enterprise documents, reducing LLM hallucinations.
- Added an agentic orchestration layer that autonomously routes between vector search and keyword fallback based on query complexity.
- Engineered a PySpark vector-ingestion pipeline (chunk, embed, index into FAISS / Azure AI Search) with incremental Delta Lake updates, plus a monitoring module logging retrieval relevance and flagging low-confidence responses.

Education

University of Cincinnati **Aug 2025 – Dec 2026**
Master of Science in Information Technology

- **Relevant Coursework:** Software Engineering, Human-Computer Interaction, Web Development, Database Systems, Cloud Computing, Data Engineering

Anna University **July 2016 – Nov 2020**
Bachelor of Engineering in Electronics and Communications Engineering

- **Relevant Coursework:** Programming in C, C++, Java, Python, Data structures and Algorithms, Big data